$$y_{ijkr} = \overbrace{\mu + \alpha_i + \beta_j + (\alpha\beta)_{ij}}^{\text{fixed effects}} + \overbrace{s_k + (\alpha s)_{ik} + (\beta s)_{jk} + (\alpha\beta s)_{ijk}}^{\text{random effects}} + \epsilon_{ijkr} \qquad \text{for } i = 1,\cdots,a;\ j = 1,\cdots,b;\ k = 1,\cdots,n;\ r = 1,\cdots,R_{ijk}\,.$$

$$\epsilon_{ijkr} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\ \sigma_\epsilon^2) \qquad s_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\ \sigma_s^2)$$

$$(\alpha s)_{ik} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\ \sigma_{\alpha s}^2) \qquad (\beta s)_{jk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\ \sigma_{\beta s}^2) \qquad (\alpha\beta s)_{ijk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\ \sigma_{\alpha\beta s}^2)$$

$$\text{Cov}(y_{ijkr},\ y_{lmkr}) = \text{Cov}\left(s_k + (\alpha s)_{ik} + (\beta s)_{jk} + (\alpha\beta s)_{ijk} + \epsilon_{ijkr},\quad s_k + (\alpha s)_{lk} + (\beta s)_{mk} + (\alpha\beta s)_{lmk} + \epsilon_{lmkr}\right)$$

$$\begin{cases} \text{Var}(y_{ij}) = \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2, & \\ \text{Cov}(y_{ij},\ y_{im}) = \sigma_s^2 + \sigma_{\alpha s}^2, & j \neq m, \\ \text{Cov}(y_{ij},\ y_{lj}) = \sigma_s^2 + \sigma_{\beta s}^2, & i \neq l, \\ \text{Cov}(y_{ij},\ y_{lm}) = \sigma_s^2, & i \neq l,\ j \neq m\,. \end{cases}$$

$$\mathbf{\Sigma}_{2\times 2,\ 2\times 2} = \begin{pmatrix} \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2 & \sigma_s^2 + \sigma_{\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 & \sigma_s^2 \\ \sigma_s^2 + \sigma_{\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2 & \sigma_s^2 & \sigma_s^2 + \sigma_{\alpha s}^2 \\ \sigma_s^2 + \sigma_{\alpha s}^2 & \sigma_s^2 & \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2 & \sigma_s^2 + \sigma_{\beta s}^2 \\ \sigma_s^2 & \sigma_s^2 + \sigma_{\alpha s}^2 & \sigma_s^2 + \sigma_{\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2 \end{pmatrix}$$

$$y_{ijkr} = \overbrace{\mu + \alpha_i + \beta_j + (\alpha\beta)_{ij}}^{\text{fixed effects}} + \overbrace{s_k + (\alpha s)_{ik} + (\beta s)_{jk} + (\alpha\beta s)_{ijk}}^{\text{random effects}} + \epsilon_{ijkr} \qquad \text{for } i = 1,\cdots,a;\ j = 1,\cdots,b;\ k = 1,\cdots,n;\ r = 1,\cdots,R_{ijk}\,.$$

$$\epsilon_{ijkr} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\ \sigma_\epsilon^2) \qquad s_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\ \sigma_s^2)$$

$$\sum_{i=1}^{a} \alpha_i = 0 \qquad \sum_{j=1}^{b} \beta_j = 0 \qquad \sum_{i=1}^{a} (\alpha\beta)_{ij} = \sum_{j=1}^{b} (\alpha\beta)_{ij} = 0$$

$$(\alpha s)_{ik} \sim \mathcal{N}(0,\ \sigma_{\alpha s}^2) \qquad \text{with } \sum_{i=1}^{a} (\alpha s)_{ik} = 0$$

$$(\beta s)_{jk} \sim \mathcal{N}(0,\ \sigma_{\beta s}^2) \qquad \text{with } \sum_{j=1}^{b} (\beta s)_{jk} = 0$$

$$(\alpha\beta s)_{ijk} \sim \mathcal{N}(0,\ \sigma_{\alpha\beta s}^2) \quad \text{with } \sum_{i=1}^{a} (\alpha\beta s)_{ijk} = \sum_{j=1}^{b} (\alpha\beta s)_{ijk} = 0$$

$$y_{11kr} = \mu - \alpha - \beta + (\alpha\beta) + s_k - (\alpha s)_k - (\beta s)_k + (\alpha\beta s)_k + \epsilon_{11kr}$$

$$y_{21kr} = \mu + \alpha - \beta - (\alpha\beta) + s_k + (\alpha s)_k - (\beta s)_k - (\alpha\beta s)_k + \epsilon_{21kr}$$

$$y_{12kr} = \mu - \alpha + \beta - (\alpha\beta) + s_k - (\alpha s)_k + (\beta s)_k - (\alpha\beta s)_k + \epsilon_{12kr}$$

$$y_{22kr} = \mu + \alpha + \beta + (\alpha\beta) + s_k + (\alpha s)_k + (\beta s)_k + (\alpha\beta s)_k + \epsilon_{22kr} \qquad \text{for } a = b = 2$$

$$\boldsymbol{\Sigma}_{2\times2,\ 2\times2} = \begin{pmatrix} \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2 & \sigma_s^2 - \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 - \sigma_{\alpha\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2 - \sigma_{\alpha\beta s}^2 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 \\ \sigma_s^2 - \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 - \sigma_{\alpha\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2 - \sigma_{\alpha\beta s}^2 \\ \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2 - \sigma_{\alpha\beta s}^2 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2 & \sigma_s^2 - \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 - \sigma_{\alpha\beta s}^2 \\ \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2 - \sigma_{\alpha\beta s}^2 & \sigma_s^2 - \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 - \sigma_{\alpha\beta s}^2 & \sigma_s^2 + \sigma_{\alpha s}^2 + \sigma_{\beta s}^2 + \sigma_{\alpha\beta s}^2 + \sigma_\epsilon^2 \end{pmatrix}$$

When $\sigma^2_{\alpha s} = \sigma^2_{\beta s} = \sigma^2_{\alpha\beta s} = \sigma^2$ in an MRE,

$$\mathbf{\Sigma}_{2\times2,\ 2\times2} = \begin{pmatrix} \sigma^2_s + 3\sigma^2 + \sigma^2_\epsilon & \sigma^2_s - \sigma^2 & \sigma^2_s - \sigma^2 & \sigma^2_s - \sigma^2 \\ \sigma^2_s - \sigma^2 & \sigma^2_s + 3\sigma^2 + \sigma^2_\epsilon & \sigma^2_s - \sigma^2 & \sigma^2_s - \sigma^2 \\ \sigma^2_s - \sigma^2 & \sigma^2_s - \sigma^2 & \sigma^2_s + 3\sigma^2 + \sigma^2_\epsilon & \sigma^2_s - \sigma^2 \\ \sigma^2_s - \sigma^2 & \sigma^2_s - \sigma^2 & \sigma^2_s - \sigma^2 & \sigma^2_s + 3\sigma^2 + \sigma^2_\epsilon \end{pmatrix} = \begin{pmatrix} \lambda_1 + \lambda_2 & \lambda_2 & \lambda_2 & \lambda_2 \\ \lambda_2 & \lambda_1 + \lambda_2 & \lambda_2 & \lambda_2 \\ \lambda_2 & \lambda_2 & \lambda_1 + \lambda_2 & \lambda_2 \\ \lambda_2 & \lambda_2 & \lambda_2 & \lambda_1 + \lambda_2 \end{pmatrix}$$

$\lambda_2 = \sigma^2_s - \sigma^2$ and $\lambda_1 = \sigma^2_\epsilon + 4\sigma^2$

Claim: Once all three random-slope variances coincide, the marginal structure has only **two** identifiable parameters.

Are two marginal variance-covariance matrices equivalent ❓ ↗️↘️ Nay. λ_2 can be negative.

Recall an RIO,

$$\mathbf{\Sigma}_{2\times2,\ 2\times2} = \begin{pmatrix} \sigma^2_\epsilon + \sigma^2_s & \sigma^2_s & \sigma^2_s & \sigma^2_s \\ \sigma^2_s & \sigma^2_\epsilon + \sigma^2_s & \sigma^2_s & \sigma^2_s \\ \sigma^2_s & \sigma^2_s & \sigma^2_\epsilon + \sigma^2_s & \sigma^2_s \\ \sigma^2_s & \sigma^2_s & \sigma^2_s & \sigma^2_\epsilon + \sigma^2_s \end{pmatrix}$$

$b = 3$

$$(d_1, d_2, d_3)^\top = \begin{pmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \\ C_{13} & C_{23} \end{pmatrix} \cdot (t_1, t_2)^\top$$

$$t_1 \perp\!\!\!\perp t_2 \sim \mathcal{N}(0, \sigma_t^2) \quad \Longrightarrow \operatorname{Var}\left((d_1, d_2, d_3)^\top\right) = \sigma_t^2 \cdot \begin{pmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \\ C_{13} & C_{23} \end{pmatrix} \cdot \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \end{pmatrix}$$

- contrasts $\sum_{i=1}^{3} C_{1i} = 0$ and $\sum_{i=1}^{3} C_{2i} = 0 \quad \Longrightarrow d_1 + d_2 + d_3 = 0$
- orthogonal contrasts $\sum_{i=1}^{3} C_{1i} C_{2i} = 0$ e.g., $\begin{pmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \\ C_{13} & C_{23} \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -1 & 1 \\ -1 & -1 \end{pmatrix}$ not unique

  orthonormal contrasts $\sum_{i=1}^{3} C_{1i}^2 = 1$ and $\sum_{i=1}^{3} C_{2i}^2 = 1$
- $C_{11}^2 + C_{21}^2 = C_{12}^2 + C_{22}^2 = C_{13}^2 + C_{23}^2 \quad \Longrightarrow \operatorname{Var}(d_1) = \operatorname{Var}(d_2) = \operatorname{Var}(d_3)$

$$\mathbf{Q}_3 = \begin{pmatrix} C_{11} & C_{21} \\ C_{12} & C_{22} \\ C_{13} & C_{23} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{6}}{3} & 0 \\ -\frac{\sqrt{6}}{6} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{6}}{6} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

Helmert (still not unique due to rotations)

$$\mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

1. $y_{11kr} = \mu - \alpha + \frac{\sqrt{6}}{3}\beta_1 - \frac{\sqrt{6}}{3}(\alpha\beta_1) + s_k - (\alpha s)_k + \frac{\sqrt{6}}{3}(\beta_1 s)_k - \frac{\sqrt{6}}{3}(\alpha\beta_1 s)_k + \epsilon_{11kr}$

2. $y_{21kr} = \mu + \alpha + \frac{\sqrt{6}}{3}\beta_1 + \frac{\sqrt{6}}{3}(\alpha\beta_1) + s_k + (\alpha s)_k + \frac{\sqrt{6}}{3}(\beta_1 s)_k + \frac{\sqrt{6}}{3}(\alpha\beta_1 s)_k + \epsilon_{21kr}$

3. $y_{12kr} = \mu - \alpha - \frac{\sqrt{6}}{6}\beta_1 + \frac{\sqrt{2}}{2}\beta_2 + \frac{\sqrt{6}}{6}(\alpha\beta_1) - \frac{\sqrt{2}}{2}(\alpha\beta_2) + s_k - (\alpha s)_k - \frac{\sqrt{6}}{6}(\beta_1 s)_k + \frac{\sqrt{2}}{2}(\beta_2 s)_k + \frac{\sqrt{6}}{6}(\alpha\beta_1 s)_k - \frac{\sqrt{2}}{2}(\alpha\beta_2 s)_k + \epsilon_{12kr}$

4. $y_{22kr} = \mu + \alpha - \frac{\sqrt{6}}{6}\beta_1 + \frac{\sqrt{2}}{2}\beta_2 - \frac{\sqrt{6}}{6}(\alpha\beta_1) + \frac{\sqrt{2}}{2}(\alpha\beta_2) + s_k + (\alpha s)_k - \frac{\sqrt{6}}{6}(\beta_1 s)_k + \frac{\sqrt{2}}{2}(\beta_2 s)_k - \frac{\sqrt{6}}{6}(\alpha\beta_1 s)_k + \frac{\sqrt{2}}{2}(\alpha\beta_2 s)_k + \epsilon_{22kr}$

5. $y_{13kr} = \mu - \alpha - \frac{\sqrt{6}}{6}\beta_1 - \frac{\sqrt{2}}{2}\beta_2 + \frac{\sqrt{6}}{6}(\alpha\beta_1) + \frac{\sqrt{2}}{2}(\alpha\beta_2) + s_k - (\alpha s)_k - \frac{\sqrt{6}}{6}(\beta_1 s)_k - \frac{\sqrt{2}}{2}(\beta_2 s)_k + \frac{\sqrt{6}}{6}(\alpha\beta_1 s)_k + \frac{\sqrt{2}}{2}(\alpha\beta_2 s)_k + \epsilon_{13kr}$

6. $y_{23kr} = \mu + \alpha - \frac{\sqrt{6}}{6}\beta_1 - \frac{\sqrt{2}}{2}\beta_2 - \frac{\sqrt{6}}{6}(\alpha\beta_1) - \frac{\sqrt{2}}{2}(\alpha\beta_2) + s_k + (\alpha s)_k - \frac{\sqrt{6}}{6}(\beta_1 s)_k - \frac{\sqrt{2}}{2}(\beta_2 s)_k - \frac{\sqrt{6}}{6}(\alpha\beta_1 s)_k - \frac{\sqrt{2}}{2}(\alpha\beta_2 s)_k + \epsilon_{23kr}$

$\mathbf{\Sigma}_{2\times3,\ 2\times3} =$

$$\begin{pmatrix}
+ & \sigma_s^2 - \sigma_{\alpha s}^2 + 2\sigma_{\beta s}^2/3 - 2\sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & 1. \\
\bullet & + & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & 2. \\
\bullet & \bullet & + & \sigma_s^2 - \sigma_{\alpha s}^2 + 2\sigma_{\beta s}^2/3 - 2\sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & 3. \\
\bullet & \bullet & \bullet & + & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & 4. \\
\bullet & \bullet & \bullet & \bullet & + & \sigma_s^2 - \sigma_{\alpha s}^2 + 2\sigma_{\beta s}^2/3 - 2\sigma_{\alpha\beta s}^2/3 & 5. \\
\bullet & \bullet & \bullet & \bullet & \bullet & \sigma_s^2 + \sigma_{\alpha s}^2 + 2\sigma_{\beta s}^2/3 + 2\sigma_{\alpha\beta s}^2/3 + \sigma_\epsilon^2 & 6. \\
1. & 2. & 3. & 4. & 5. & 6. &
\end{pmatrix}$$

Variance of the difference scores for factor B

$$\mathrm{Var}(y_{i1} - y_{i2}) = \mathrm{Var}(y_{i1}) + \mathrm{Var}(y_{i2}) - 2\,\mathrm{Cov}(y_{i1},\, y_{i2}) = 2\sigma_{\beta s}^2 + 2\sigma_{\alpha\beta s}^2 + 2\sigma_\epsilon^2$$

$$\mathrm{Var}(y_{i1} - y_{i3}) = \mathrm{Var}(y_{i1}) + \mathrm{Var}(y_{i3}) - 2\,\mathrm{Cov}(y_{i1},\, y_{i3}) = 2\sigma_{\beta s}^2 + 2\sigma_{\alpha\beta s}^2 + 2\sigma_\epsilon^2$$

$$\mathrm{Var}(y_{i2} - y_{i3}) = \mathrm{Var}(y_{i2}) + \mathrm{Var}(y_{i3}) - 2\,\mathrm{Cov}(y_{i2},\, y_{i3}) = 2\sigma_{\beta s}^2 + 2\sigma_{\alpha\beta s}^2 + 2\sigma_\epsilon^2$$

1. $y_{11kr} = \mu - \alpha + \frac{\sqrt{6}}{3}\beta_1 - \frac{\sqrt{6}}{3}(\alpha\beta_1) + s_k - (\alpha s)_k + \frac{\sqrt{6}}{3}(\beta_1 s)_k - \frac{\sqrt{6}}{3}(\alpha\beta_1 s)_k + \epsilon_{11kr}$

2. $y_{21kr} = \mu + \alpha + \frac{\sqrt{6}}{3}\beta_1 + \frac{\sqrt{6}}{3}(\alpha\beta_1) + s_k + (\alpha s)_k + \frac{\sqrt{6}}{3}(\beta_1 s)_k + \frac{\sqrt{6}}{3}(\alpha\beta_1 s)_k + \epsilon_{21kr}$

3. $y_{12kr} = \mu - \alpha - \frac{\sqrt{6}}{6}\beta_1 + \frac{\sqrt{2}}{2}\beta_2 + \frac{\sqrt{6}}{6}(\alpha\beta_1) - \frac{\sqrt{2}}{2}(\alpha\beta_2) + s_k - (\alpha s)_k - \frac{\sqrt{6}}{6}(\beta_1 s)_k + \frac{\sqrt{2}}{2}(\beta_2 s)_k + \frac{\sqrt{6}}{6}(\alpha\beta_1 s)_k - \frac{\sqrt{2}}{2}(\alpha\beta_2 s)_k + \epsilon_{12kr}$

4. $y_{22kr} = \mu + \alpha - \frac{\sqrt{6}}{6}\beta_1 + \frac{\sqrt{2}}{2}\beta_2 - \frac{\sqrt{6}}{6}(\alpha\beta_1) + \frac{\sqrt{2}}{2}(\alpha\beta_2) + s_k + (\alpha s)_k - \frac{\sqrt{6}}{6}(\beta_1 s)_k + \frac{\sqrt{2}}{2}(\beta_2 s)_k - \frac{\sqrt{6}}{6}(\alpha\beta_1 s)_k + \frac{\sqrt{2}}{2}(\alpha\beta_2 s)_k + \epsilon_{22kr}$

5. $y_{13kr} = \mu - \alpha - \frac{\sqrt{6}}{6}\beta_1 - \frac{\sqrt{2}}{2}\beta_2 + \frac{\sqrt{6}}{6}(\alpha\beta_1) + \frac{\sqrt{2}}{2}(\alpha\beta_2) + s_k - (\alpha s)_k - \frac{\sqrt{6}}{6}(\beta_1 s)_k - \frac{\sqrt{2}}{2}(\beta_2 s)_k + \frac{\sqrt{6}}{6}(\alpha\beta_1 s)_k + \frac{\sqrt{2}}{2}(\alpha\beta_2 s)_k + \epsilon_{13kr}$

6. $y_{23kr} = \mu + \alpha - \frac{\sqrt{6}}{6}\beta_1 - \frac{\sqrt{2}}{2}\beta_2 - \frac{\sqrt{6}}{6}(\alpha\beta_1) - \frac{\sqrt{2}}{2}(\alpha\beta_2) + s_k + (\alpha s)_k - \frac{\sqrt{6}}{6}(\beta_1 s)_k - \frac{\sqrt{2}}{2}(\beta_2 s)_k - \frac{\sqrt{6}}{6}(\alpha\beta_1 s)_k - \frac{\sqrt{2}}{2}(\alpha\beta_2 s)_k + \epsilon_{23kr}$

$\mathbf{\Sigma}_{2\times3,\ 2\times3} =$

$$\begin{pmatrix}
+ & \sigma_s^2 - \sigma_{\alpha s}^2 + 2\sigma_{\beta s}^2/3 - 2\sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & 1. \\
\bullet & + & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & 2. \\
\bullet & \bullet & + & \sigma_s^2 - \sigma_{\alpha s}^2 + 2\sigma_{\beta s}^2/3 - 2\sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & 3. \\
\bullet & \bullet & \bullet & + & \sigma_s^2 - \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 + \sigma_{\alpha\beta s}^2/3 & \sigma_s^2 + \sigma_{\alpha s}^2 - \sigma_{\beta s}^2/3 - \sigma_{\alpha\beta s}^2/3 & 4. \\
\bullet & \bullet & \bullet & \bullet & + & \sigma_s^2 - \sigma_{\alpha s}^2 + 2\sigma_{\beta s}^2/3 - 2\sigma_{\alpha\beta s}^2/3 & 5. \\
\bullet & \bullet & \bullet & \bullet & \bullet & \sigma_s^2 + \sigma_{\alpha s}^2 + 2\sigma_{\beta s}^2/3 + 2\sigma_{\alpha\beta s}^2/3 + \sigma_\epsilon^2 & 6.
\end{pmatrix}$$

1. 2. 3. 4. 5. 6.

Variance of the difference scores

for the interaction *AB*

$$\text{Var}\left((y_{11} - y_{21}) - (y_{12} - y_{22})\right) = 8\sigma_{\alpha\beta s}^2 + 4\sigma_\epsilon^2$$

$$\text{Var}\left((y_{11} - y_{21}) - (y_{13} - y_{23})\right) = 8\sigma_{\alpha\beta s}^2 + 4\sigma_\epsilon^2$$

$$\text{Var}\left((y_{12} - y_{22}) - (y_{13} - y_{23})\right) = 8\sigma_{\alpha\beta s}^2 + 4\sigma_\epsilon^2$$

$$
\begin{aligned}
\boldsymbol{\Sigma}_{ab} &= \sigma^2_\epsilon \mathbf{I}_{ab} + \sigma^2_s \mathbf{J}_{ab} + \sigma^2_{\alpha s} \mathbf{J}_b \otimes \left(\mathbf{Q}_a \mathbf{Q}_a^\top\right) + \sigma^2_{\beta s}\left(\mathbf{Q}_b \mathbf{Q}_b^\top\right) \otimes \mathbf{J}_a + \sigma^2_{\alpha\beta s}\left(\mathbf{Q}_b \mathbf{Q}_b^\top\right) \otimes \left(\mathbf{Q}_a \mathbf{Q}_a^\top\right) \\
&= \sigma^2_\epsilon \mathbf{I}_{ab} + \sigma^2_s \mathbf{J}_{ab} + \sigma^2_{\alpha s} \mathbf{J}_b \otimes \left(\mathbf{I}_a - a^{-1}\mathbf{J}_a\right) + \sigma^2_{\beta s}\left(\mathbf{I}_b - b^{-1}\mathbf{J}_b\right) \otimes \mathbf{J}_a + \sigma^2_{\alpha\beta s}\left(\mathbf{I}_b - b^{-1}\mathbf{J}_b\right) \otimes \left(\mathbf{I}_a - a^{-1}\mathbf{J}_a\right) \\
&= \sigma^2_\epsilon \mathbf{I}_{ab} + \sigma^2_s \mathbf{J}_{ab} + \sigma^2_{\alpha s}\left(\mathbf{J}_b \otimes \mathbf{I}_a - a^{-1}\mathbf{J}_{ab}\right) + \sigma^2_{\beta s}\left(\mathbf{I}_b \otimes \mathbf{J}_a - b^{-1}\mathbf{J}_{ab}\right) + \sigma^2_{\alpha\beta s}\left(\mathbf{I}_{ab} - b^{-1}\mathbf{J}_b \otimes \mathbf{I}_a - a^{-1}\mathbf{I}_b \otimes \mathbf{J}_a + (ab)^{-1}\mathbf{J}_{ab}\right) \\
&= \text{sum of compound symmetry matrices } + \ (\sigma^2_{\alpha s} - b^{-1}\sigma^2_{\alpha\beta s})\mathbf{J}_b \otimes \mathbf{I}_a + (\sigma^2_{\beta s} - a^{-1}\sigma^2_{\alpha\beta s})\mathbf{I}_b \otimes \mathbf{J}_a
\end{aligned}
$$

The remaining part exhibits compound symmetry if and only if
$\sigma^2_{\alpha s} - b^{-1}\sigma^2_{\alpha\beta s} = \sigma^2_{\beta s} - a^{-1}\sigma^2_{\alpha\beta s} = 0$.

If $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{B}$ is a $p \times q$ matrix, then the Kronecker product $\mathbf{A} \otimes \mathbf{B}$ is the $pm \times qn$ block matrix:

$$
\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & \ddots & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{pmatrix}
$$

$$x_2\mathbf{J}_3 \otimes \mathbf{I}_3 = \left(\begin{array}{ccc:ccc:ccc} x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 & 0 & 0 \\ 0 & x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 & 0 \\ 0 & 0 & x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 \\ \hdashline x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 & 0 & 0 \\ 0 & x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 & 0 \\ 0 & 0 & x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 \\ \hdashline x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 & 0 & 0 \\ 0 & x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 & 0 \\ 0 & 0 & x_2 & 0 & 0 & x_2 & 0 & 0 & x_2 \end{array}\right)$$

$$x_1\mathbf{I}_3 \otimes \mathbf{J}_3 = \left(\begin{array}{ccc:ccc:ccc} x_1 & x_1 & x_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ x_1 & x_1 & x_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ x_1 & x_1 & x_1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hdashline 0 & 0 & 0 & x_1 & x_1 & x_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_1 & x_1 & x_1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_1 & x_1 & x_1 & 0 & 0 & 0 \\ \hdashline 0 & 0 & 0 & 0 & 0 & 0 & x_1 & x_1 & x_1 \\ 0 & 0 & 0 & 0 & 0 & 0 & x_1 & x_1 & x_1 \\ 0 & 0 & 0 & 0 & 0 & 0 & x_1 & x_1 & x_1 \end{array}\right)$$

Sphericity (Circularity)

The variances of the differences between any pair of within-participant conditions are equal.

Compound Symmetry

All conditions have equal variance and all pairs of conditions have equal covariance.

Sphericity

- The variances of all pairwise differences between repeated measures are equal.
- The orthogonal contrasts have equal variances and zero covariances.

The second definition generalizes more naturally to interaction effects in multiway repeated-measures designs.

Langenberg, B., Helm, J. L., Günther, T., & Mayer, A. (2023). Understanding, testing, and relaxing sphericity of repeated measures ANOVA with manifest and latent variables using SEM. *Methodology*, *19*, 60–95. https://doi.org/10.5964/meth.8415

Lane, D. M. (2016). The assumption of sphericity in repeated-measures designs: What it means and what to do when it is violated. *Quantitative Methods for Psychology*, *12*, 114–122. https://doi.org/10.20982/tqmp.12.2.p114

Note: In multivariate analysis, sphericity refers to the covariance matrix being proportional to the identity matrix, i.e., $\sigma^2\mathbf{I}$.

Fisher, T. J., Sun, X., & Gallagher, C. M. (2010). A new test for sphericity of the covariance matrix for high dimensional data. *Journal of Multivariate Analysis*, *101*, 2554–2570. https://doi.org/10.1016/j.jmva.2010.07.004